Ahmed Mohamed Samir

Mechatronics Engineer

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PROFESSIONAL SUMMARY

Motivated Mechatronics Engineering graduate with comprehensive knowledge in mechanical design, embedded systems, control systems, robotics, and industrial automation. Proven ability to integrate mechanics, electronics, and programming to develop innovative and efficient engineering solutions. Experienced in real-world mechanical engineering projects, maintenance, and quality control within construction, railway, and petroleum industries. Skilled in PLC, SCADA, HMI, and industrial communication with strong problem-solving and leadership capabilities.

EDUCATION

Ain Shams University — Bachelor of Mechanical Engineering (Minor: Mechatronics)
Expected Graduation: June 2025

PROFESSIONAL EXPERIENCE

Cheiron Petroleum Corporation — Intern
2024

- Gained hands-on experience with mechanical systems and equipment in oil field operations.
- Assisted in maintenance of pumps, compressors, and piping systems, enhancing understanding of mechanical engineering in the oil & gas sector.

Egyptian National Railways — Intern
2023

- Observed and supported scheduled maintenance and mechanical repairs on various train engine types.
- Learned operation and control systems of railway locomotives, improving knowledge of transport engineering and heavy machinery.

Arab Contractors Company — Intern
2022

- Assisted engineers in daily site activities and project coordination for mechanical engineering tasks.
- Supported project management and quality control practices, strengthening practical fieldwork skills.

PROJECTS

Graduation Project:

- Developed an intelligent, fully automated testing machine for prosthetic feet and footwear durability and performance evaluation.
- Integrated advanced control systems, smart monitoring, and IoT connectivity, adhering to ISO international standards to ensure quality of prosthetics.

Robotic Arm Project:

- Designed, assembled, and programmed a robotic arm integrating mechanical structure, electronics, and motion control.

Water Turbine Project:

- Designed and operated a small-scale water turbine generating mechanical and electrical power.

RC Line Follower Car:

- Built an autonomous RC car using sensors and control algorithms to follow a predefined path.

Conveyor Control Panel:

- Designed an electrical control panel to efficiently operate and manage conveyor belt functions.

CERTIFICATIONS

- Water Turbine Design
- Ultimate Electrical Fundamentals
- SolidWorks
- Financial Account Fundamentals

SKILLS

- Mechanical Design
- Embedded Systems
- Control Systems
- Robotics
- Industrial Automation
- PLC, SCADA, HMI
- Industrial Communication
- Leadership
- MEP (HVAC, Firefighting)
- Microsoft Office Suite
- Programming: Python, C++
- Problem Solving
- Teamwork

LANGUAGES

- Arabic (Native)
- English (Very Good)